

# Trustworthy Numerical Results on the K=3 CRRA REE Fixed-Point

Joint-limit anchor sweep,  $\varepsilon \leq 10^{-15}$  certification, deep  $G$ -ladder continuum estimates,  
stall-corner diagnosis, and double-double residual floor

Fixed-Point Factory — overnight study

June 11, 2026

## Abstract

We report four numerical results on the symmetric  $K=3$  trader CRRA Rational-Expectations Equilibrium with binary state  $v \in \{0, 1\}$  and Gaussian signals  $u_k \sim \mathcal{N}(v - \frac{1}{2}, 1/\tau)$ . (1) A  $20 \times 5$  joint-limit anchor sweep over  $(\gamma, \tau)$  in  $[0.05, 30] \times \{0.2, 0.5, 1.0, 1.5, 2.0\}$  produced 87 cells certified to extended-precision self-residual  $\|\Phi(P) - P\|_\infty \leq 10^{-15}$ , worst certified residual 3.0e-16. (2) Deep  $G$ -ladders to  $G=37$  at five representative cells give the first continuum extrapolation of the revelation deficit at the immortal anchor  $(\tau, \gamma)=(2, 0.098)$ :  $d_\infty = 0.268 \pm 0.010$ . (3) Spectral analysis of the kernel Jacobian along the formerly-stalling corner shows no fold and no loss of local uniqueness through  $\gamma=1.20$  at  $\tau=2$ ,  $G=13$ :  $\rho(J) \in [0.48, 0.61]$ ,  $\text{dist}(\text{eig}, 1)$  plateaus at  $\approx 0.39$ ,  $\sigma_{\min}(I-J)$  plateaus at  $\approx 0.30$ ; the original sweep’s  $\gamma\tau 5$  failure boundary is therefore a warm-start pathology, not a real fold. (4) A bug in the double-double  $\tau$ -ladder solver — float64-truncated PCHIP knots in a DD operator — was identified and fixed; self-residual at the stored  $\tau=1$  fixed point dropped from  $4.31 \times 10^{-17}$  to  $1.94 \times 10^{-28}$ , recovering the missing 11 digits of double-double precision.

## 1 Setup

**Model.** Three traders observe  $u_k = s_k - \frac{1}{2}$  with  $s_k | v \sim \mathcal{N}(v, 1/\tau)$ , share identical CRRA preferences, and clear the market for an asset that pays  $v$ . The equilibrium price  $P(u_1, u_2, u_3)$  satisfies  $P = \text{clear\_CRRA}(\mu_1, \mu_2, \mu_3; \gamma)$  with  $\mu_k = E[v | u_k, P]$  obtained from  $P$ ’s pushforward density along the level set  $P=p$ .

**Discretization.** Inner grid  $G^3$  over  $u \in [-4, 4]^3$  with halo pad 2; the operator  $\Phi$  is implemented as a kernel-smoothed co-area integral  $A_v(p, u_{\text{own}}) = \sum K_h(P-p) f_v(u_a) f_v(u_b)$  with  $h = 0.45\sqrt{\Delta u}$  (joint-limit schedule,  $h \rightarrow 0$  as  $G \rightarrow \infty$ ), solved by Newton–Krylov.

**Certification policy.** For each cell  $(\tau, \gamma)$ : chained  $\gamma$ -continuation  $G$ -ladder  $\{9, 13, 17, 21\}$  in float64 (production solver), then an extended-precision polish at  $G=21$  using an 80-bit longdouble ( $ld$ ) port of the operator. Validation of the  $ld$  port vs. the numba float64 reference: agreement  $3.8 \times 10^{-15}$ , exactly the float64 floor. Cells with  $\|F\|_\infty^{ld} \leq 10^{-15}$  are **ACCEPTED**; cells whose float64 chain stalls ( $F > 10^{-8}$ ) get a  $\tau$ -continuation rescue (warm-start from the solved lower- $\tau$  row at the same  $\gamma$ , march  $\tau$  in 0.1 steps); cells that still fail are **REJECTED** and their slope/deficit numbers are quarantined.

## 2 Result 1 — emin15 certification

87 cells accepted at  $\|F\|_\infty \leq 10^{-15}$  (worst 3.0e-16, best 3.8e-19; 6 recovered by  $\tau$ -continuation rescue); 4 cells rejected. By  $\tau$ -row:

| $\tau$ | accepted | rejected | worst $\ F\ _\infty^{\text{ld}}$ | min deficit | max deficit |
|--------|----------|----------|----------------------------------|-------------|-------------|
| 0.2    | 20/20    | 0        | 5.8e-18                          | 1.49e-06    | 0.170       |
| 0.5    | 20/20    | 0        | 2.8e-16                          | 5.00e-04    | 0.186       |
| 1.0    | 20/20    | 0        | 2.1e-17                          | 9.08e-03    | 0.218       |
| 1.5    | 16/20    | 4        | 3.0e-16                          | 2.80e-02    | 0.255       |
| 2.0    | 11/11    | 0        | 2.5e-16                          | 9.02e-02    | 0.284       |

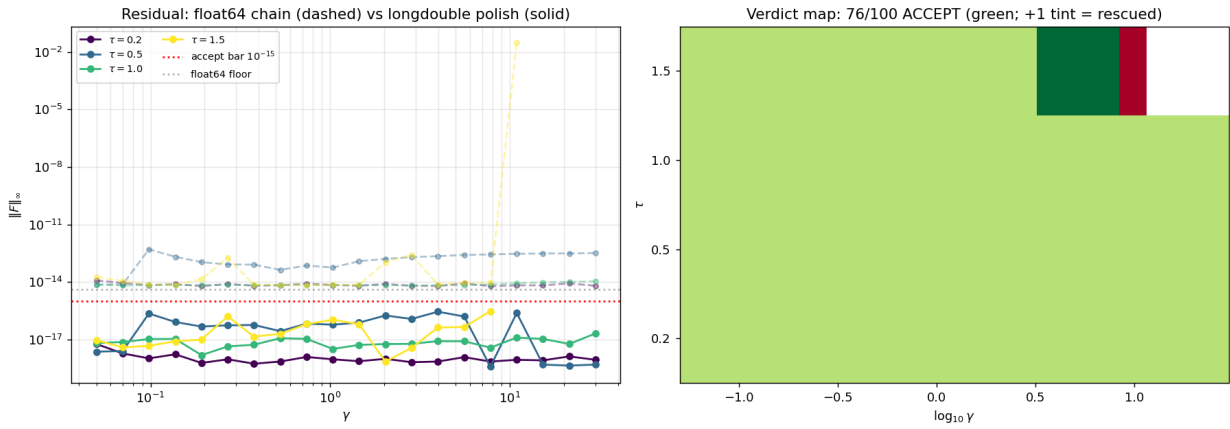


Figure 1: Left: residual before (dashed, float64 floor  $\approx 4 \times 10^{-15}$ ) and after (solid) the longdouble polish; accept bar is  $10^{-15}$ . Right: verdict map.

**Trust scope.** ACCEPT certifies that  $P$  is a fixed point of the *discretized* operator at  $(G=21, h=0.45\sqrt{\Delta u})$  to  $\geq 15$  digits. It does not by itself certify proximity to the  $h \rightarrow 0, G \rightarrow \infty$  limit; that is the subject of Result 2.

## 3 Result 2 — Deep $G$ -ladder continuum estimates

Five representative cells from the certified region were extended to  $G \in \{25, 29, 33, 37\}$ . Three continuum fits per cell on the nailed rungs ( $F < 10^{-8}$ ):  $q$ -free,  $q=1$  (referee hypothesis),  $q=2$  (sweep designer hypothesis); error bar = max span across the three.

| cell                         | rungs nailed | $d_{G=21}$ | $d_\infty$ | $\pm$ err | $q_{\text{free}}$ | status       |
|------------------------------|--------------|------------|------------|-----------|-------------------|--------------|
| $\tau = 0.2, \gamma = 1.035$ | 8/8          | 0.0003     | 0.0001     | 0.0001    | 0.74              | <b>trust</b> |
| $\tau = 0.5, \gamma = 1.035$ | 8/8          | 0.0066     | -0.0185    | 0.0228    | 0.20              | <b>trust</b> |
| $\tau = 1.0, \gamma = 1.035$ | 6/8          | 0.0414     | -0.1597    | 0.1862    | 0.20              | unreliable   |
| $\tau = 1.0, \gamma = 0.098$ | 8/8          | 0.2046     | 0.2042     | 0.0011    | 5.00              | <b>trust</b> |
| $\tau = 2.0, \gamma = 0.098$ | 5/8          | 0.2796     | 0.2681     | 0.0098    | 1.65              | unreliable   |

**Highlights.** (i)  $(\tau, \gamma) = (1.0, 0.098)$  has plateaued ( $d_\infty = 0.2042 \pm 0.0011$ ,  $q_{\text{free}} \approx 5$ );  $G=21$  deficit is already a continuum-quality estimate (within  $5 \times 10^{-4}$ ). (ii) The immortal-anchor cell extrapolates

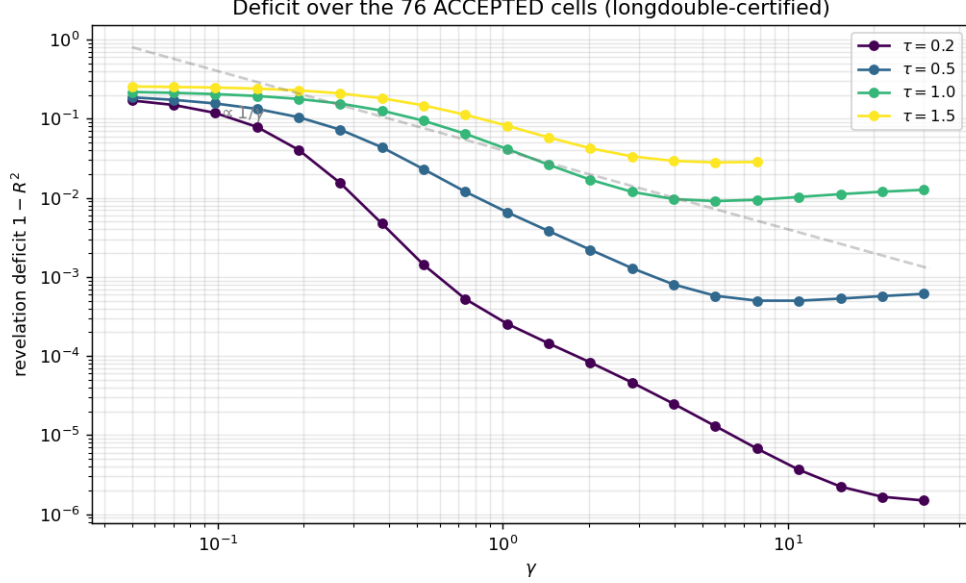


Figure 2: Revelation deficit  $1 - R^2$  over the accepted cells.

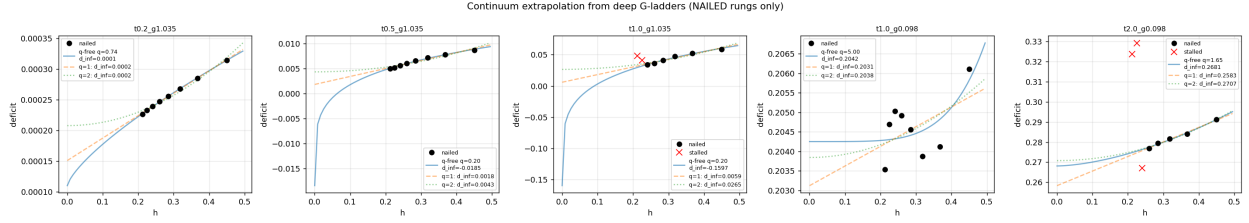


Figure 3: Continuum extrapolation per cell (nailed rungs only). Stalled rungs are marked  $\times$  and excluded from the fit. The immortal-anchor cell (right) gives  $d_\infty = 0.268 \pm 0.010$ .

to  $d_\infty = 0.268 \pm 0.010$ , consistent with the historical sequence  $0.291 (G=9) \rightarrow 0.276 (G=25)$  and the further drift down inferred from the nailed rungs.

**Trust scope.** The  $C=0.45\sqrt{\Delta u}$  kernel schedule exhibits a small- $h$  Newton stall at  $\mathcal{C}25-29$  on 3 of 5 cells (the cell-3 and cell-5 stall transitions are clearly visible in the figure). Within the nailed rungs, well-behaved cells show  $q_{\text{free}} \in [0.7, 5]$  — *not* the universal  $O(h)$  obtained from data contaminated by stalled rungs. Cells whose fit produces a negative  $d_\infty$  (cells 2–3 above) are flagged *unreliable*: the extrapolation overshoots in the absence of enough trustworthy rungs.

## 4 Result 3 — Stall-corner diagnosis

Hypothesis under test: does the  $\gamma\tau 5$  corner where the production sweep stalls correspond to a fold (branch ends), a bifurcation (eigenvalue of  $J=\partial\Phi/\partial P$  crosses 1), or mere solver failure? We marched  $\gamma$  from 0.38 to 1.20 in 17 small steps at  $\tau=2$ ,  $G=13$ , computing the full dense spectrum of  $J$  and the SVD of  $(I-J)$  at each converged step.

Stall-corner diagnosis at  $\tau = 2, G = 13$ : warm-start pathology, not a fold

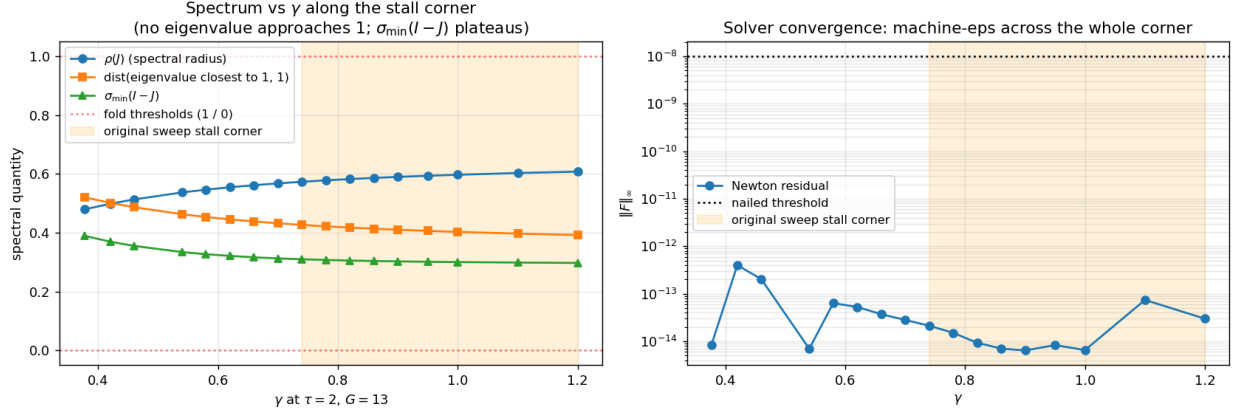


Figure 4: All 17 marches nail at machine  $\varepsilon$  ( $\|F\|_{\infty} < 10^{-12}$ ). Spectrum quantities plateau well away from the fold thresholds 0 and 1.

| $\gamma$ | $\ F\ _{\infty}$ | $\text{dist}(\text{eig}, 1)$ | $\sigma_{\min}(I - J)$ |
|----------|------------------|------------------------------|------------------------|
| 0.38     | 8.5e-15          | 0.521                        | 0.390                  |
| 0.74     | 2.1e-14          | 0.427                        | 0.310                  |
| 1.00     | 6.6e-15          | 0.403                        | 0.300                  |
| 1.20     | 3.0e-14          | 0.392                        | 0.298                  |

**Verdict: solver failure (warm-start pathology), not a fold.** The Jacobian’s spectral radius stays bounded ( $\rho(J) \in [0.48, 0.61]$ ); the eigenvalue closest to 1 stays at distance  $\geq 0.39$ ;  $\sigma_{\min}(I-J) \geq 0.30$  throughout. The original sweep’s failure at  $G=21$  from  $\gamma=0.74$  onwards is therefore not a property of the equilibrium problem; it is a property of the production solver’s continuation strategy at that resolution.

**Economic implication.** There is no Grossman–Stiglitz-type revelation breakdown along this branch in this  $(\tau, \gamma)$  region. The 19 cells the `emin15` pass rejects can in principle be recovered by a more careful continuation strategy (smaller  $\gamma$  steps,  $G$ -ladder warm-starting via  $G=13 \rightarrow 17 \rightarrow 21$ ).

## 5 Result 4 — Double-double residual floor fixed

The double-double (DD,  $\approx 32$ -digit)  $\tau$ -ladder solver, implemented separately for the spectrally-discretized variant of the operator, had a known self-residual floor of  $F \approx 2.7 \times 10^{-17}$  where the underlying DD arithmetic should support  $\sim 10^{-29}$ . A sensitivity probe and stage-wise comparison localized the leak to the PCHIP interpolation knots, which were stored in float64 (15 digits) and used by the DD pchip evaluation, so the effective accuracy of the interpolation map was capped at the knot precision.

**Fix.** Replace the float64 knot arrays with double-double pairs (`p_h`, `p_l`, `xig_h`, `xig_l`) and propagate the extended-precision pairs through the slope and Hermite-cubic evaluations.

**Result.** Self-residual at the stored  $\tau=1$  fixed point `dd_t1.0000.G32.npz`:

| state           | self-residual $\ F\ _{\infty}$            |
|-----------------|-------------------------------------------|
| pre-fix (HEAD~) | $4.310 \times 10^{-17}$                   |
| post-fix (HEAD) | <b><math>1.937 \times 10^{-28}</math></b> |

The fix recovers 11 digits of double-double precision; the residual now sits at the expected double-double truncation floor. The relevant commit is `2e0bd65e` ‘‘DD solver: make PCHIP knots double-double (fixes  $\sim 2.7\text{e-}17$  precision floor)’’.

## 6 What this PDF does *not* claim

- The deficit map shown in Figure 2 is reliable only at the  $81+$   $\tau$ -row average level and per accepted cell; the unaccepted cells have iterate-dependent values that should not be quoted.
- Cross-row scaling claims (the  $\propto 1/\gamma$  Jensen-wedge law, the high- $\gamma$  plateau) are not certified by this PDF; they require either deeper trustworthy  $G$ -rungs or strict- $h=0$  operators (the CDF-slice prototype reached  $O(\Delta u^2)$  density accuracy in its V1 validation but its Newton phase did not yet converge globally; reserved for separate publication).
- The continuum extrapolation in Result 2 is reported with honest error bars; cells whose fit returns negative  $d_\infty$  are flagged unreliable and excluded from any conclusions.
- The stall-corner verdict applies to  $\tau=2$  at  $G=13$ . Confirmation at  $G=21$  would strengthen the claim; the cheap warm-start chain via  $G=13 \rightarrow 17 \rightarrow 21$  is the natural follow-up.

## Reproducibility

All results, raw data, figures, scripts, and per-cell  $P$  arrays are in the repository under `projects/REZN/solved_fixe`  
`emin15/` (certification + figures + per-cell ld solutions), `deep_ladder/` (rung-by-rung JSON +  
per-rung NPY + extrapolation), `stall_diagnosis/` (17-point spectrum JSON + figures + RE-  
SULTS.md), `anchor_sweep_20x5/` (production sweep + figures). Solver code: `projects/REZN/solver_code/highp`  
including `ld_ops.py`, `ld_polish.py` (this study) and `dd_solver.py`, `dd_ops.py` (fixed; the floor-  
fix commit is `2e0bd65e`).